

Functions and Graphs — Complete Notes

Grade 11

Mathematics

CAPS

What you'll learn

- Sketch and interpret the parabola, hyperbola and exponential function.
- Find turning points, axes of symmetry, intercepts and asymptotes.
- State the domain and range of each function accurately.
- Apply vertical and horizontal shifts, reflections and stretches to a graph.
- Calculate the average gradient of a function between two points.

The vocabulary you must know

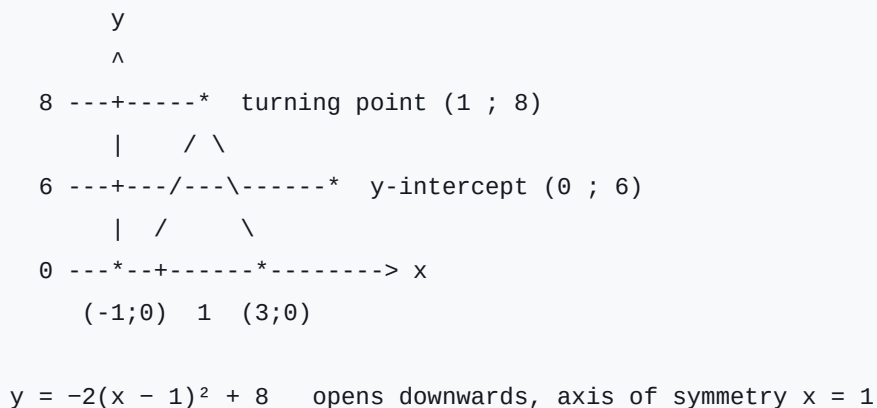
A function is a rule that gives exactly one output for every input. The set of allowed inputs is the domain, and the set of possible outputs is the range.

- x-intercept: where the graph cuts the x-axis, found by setting $y = 0$.
- y-intercept: where the graph cuts the y-axis, found by setting $x = 0$.
- Turning point: the maximum or minimum point of a parabola.
- Axis of symmetry: the vertical line through the turning point.
- Asymptote: a line that the graph gets closer and closer to but never touches.

The parabola (quadratic function)

The turning-point form of a parabola is $y = a(x - p)^2 + q$.

- Turning point: $(p ; q)$.
- Axis of symmetry: $x = p$.
- Opens upwards if $a > 0$, downwards if $a < 0$.
- Range: $y \geq q$ if $a > 0$, or $y \leq q$ if $a < 0$.



If the parabola is given in standard form $y = ax^2 + bx + c$, the x-coordinate of the turning point is $x = -b \div (2a)$. Substitute that value back into the equation to get the y-coordinate.

The hyperbola

The standard form of a hyperbola is $y = a \div (x - p) + q$.

- Vertical asymptote: $x = p$.
- Horizontal asymptote: $y = q$.
- Domain: $x \neq p$.
- Range: $y \neq q$.
- If $a > 0$ the two branches sit in the top-right and bottom-left regions; if $a < 0$ they swap.



$y = 3/(x - 2) + 1$ asymptotes at $x = 2$ and $y = 1$

The exponential function

The standard form is $y = a \times b^x + q$, where $b > 0$ and $b \neq 1$.

- Horizontal asymptote: $y = q$.
- If $b > 1$ the graph increases; if $0 < b < 1$ it decreases.
- If $a > 0$ the range is $y > q$; if $a < 0$ the range is $y < q$.
- The graph always cuts the y-axis at $y = a + q$ (put $x = 0$).



$y = 2 \times 3^x - 1$ increasing, approaches $y = -1$ but never reaches it

Transformations

Every change to the equation moves or reshapes the graph. Starting from $y = f(x)$:

Transformation	Effect on the graph
$y = f(x) + q$	shift up by q (down if q is negative)
$y = f(x - p)$	shift right by p (left if p is negative)
$y = -f(x)$	reflect in the x -axis
$y = f(-x)$	reflect in the y -axis
$y = a \cdot f(x)$	vertical stretch by a factor a (or a reflection if $a < 0$)

A useful check: inside the bracket, the sign works the opposite way to what you might expect. $y = (x - 3)^2$ shifts three units to the right, not left.

Average gradient

The average gradient of a function f between $x = a$ and $x = b$ is the gradient of the straight line joining the two points on the graph:

$$\text{Average gradient} = (f(b) - f(a)) \div (b - a)$$

It measures how fast the function changes on average over that interval. For a straight line this is just the gradient of the line; for a curve it is the gradient of the secant between the two points.

Worked example 1: Sketching a parabola

Sketch $y = -2(x - 1)^2 + 8$ and state its intercepts, turning point and range.

Step 1 — Read off the shape. The equation is already in turning-point form with $a = -2$, $p = 1$, $q = 8$. Because $a < 0$ the parabola opens downwards and the turning point is a maximum.

Turning point: $(1 ; 8)$. Axis of symmetry: $x = 1$.

Step 2 — Find the y -intercept by setting $x = 0$. $y = -2(0 - 1)^2 + 8 = -2(1) + 8 = 6$. So the y -intercept is $(0 ; 6)$.

Step 3 — Find the x -intercepts by setting $y = 0$. $-2(x - 1)^2 + 8 = 0$ $-2(x - 1)^2 = -8$ $(x - 1)^2 = 4$ $x - 1 = \pm 2$ $x = 3$ or $x = -1$. So the x -intercepts are $(3 ; 0)$ and $(-1 ; 0)$.

Step 4 — State the range. The maximum value is the turning point's y -value, so $y \leq 8$.

Answer: turning point $(1 ; 8)$, x -intercepts at -1 and 3 , y -intercept at 6 , range $y \leq 8$.

Worked example 2: Analysing a hyperbola

For $y = 3/(x - 2) + 1$, write down the asymptotes, the domain, the range and the intercepts.

Step 1 — Identify the asymptotes from the form $y = a/(x - p) + q$. Vertical asymptote: $x = 2$. Horizontal asymptote: $y = 1$.

Step 2 — Domain and range. The graph is undefined at $x = 2$, so the domain is $x \neq 2$. The graph never reaches $y = 1$, so the range is $y \neq 1$.

Step 3 — y -intercept (put $x = 0$). $y = 3/(0 - 2) + 1 = 3/(-2) + 1 = -3/2 + 1 = -1/2$.

Step 4 — x -intercept (put $y = 0$). $0 = 3/(x - 2) + 1$ $-1 = 3/(x - 2)$ $-1(x - 2) = 3$ $x - 2 = -3$ $x = -1$.

Answer: asymptotes $x = 2$ and $y = 1$; domain $x \neq 2$; range $y \neq 1$; y -intercept $-1/2$; x -intercept -1 .

Worked example 3: Exponential graph and average gradient

Given $f(x) = 2 \times 3^x - 1$, state the asymptote and range, find $f(0)$, and calculate the average gradient between $x = 0$ and $x = 2$.

Step 1 — Read the asymptote and range. Here $a = 2$ (positive) and $q = -1$, so the horizontal asymptote is $y = -1$ and, because $a > 0$, the range is $y > -1$.

Step 2 — Find $f(0)$. $f(0) = 2 \times 3^0 - 1 = 2 \times 1 - 1 = 1$. So the y-intercept is $(0 ; 1)$.

Step 3 — Find $f(2)$. $f(2) = 2 \times 3^2 - 1 = 2 \times 9 - 1 = 18 - 1 = 17$.

Step 4 — Apply the average gradient formula. Average gradient $= (f(2) - f(0)) \div (2 - 0) = (17 - 1) \div 2 = 16 \div 2 = 8$.

Answer: asymptote $y = -1$; range $y > -1$; $f(0) = 1$; average gradient $= 8$.

Finding the equation from a graph

If you are given the turning point and one other point on a parabola, substitute into $y = a(x - p)^2 + q$ using the turning point for p and q , then use the other point to solve for a . For a hyperbola, read the asymptotes to get p and q , then substitute any known point to find a .

Common mistakes

- Getting the sign of p wrong. The turning point of $y = (x - 2)^2 + 5$ is $(2 ; 5)$, not $(-2 ; 5)$.
- Confusing domain and range. Domain is about the inputs (x -values), range is about the outputs (y -values).
- Forgetting to exclude the asymptote from the range of a hyperbola or exponential graph.
- Believing an exponential graph can be negative. The value 3^x is always positive, so $y = 3^x$ never dips below the x -axis.
- Reporting the average gradient where a single point's gradient was asked for, or the other way round.
- Plotting too few points and drawing a parabola as a set of straight lines instead of a smooth curve.
- Mixing up reflections: $y = -f(x)$ flips the graph vertically, while $y = f(-x)$ flips it horizontally.

Quick summary / cheat sheet

Function	Form	Key features
Parabola	$y = a(x - p)^2 + q$	turning point $(p ; q)$, axis $x = p$, opens up if $a > 0$
Hyperbola	$y = a/(x - p) + q$	asymptotes $x = p$ and $y = q$, domain $x \neq p$, range $y \neq q$
Exponential	$y = a \cdot b^x + q$	asymptote $y = q$, range $y > q$ if $a > 0$, always increasing or decreasing

Transformations of $y = f(x)$:

- $+q$ shifts up, $-q$ shifts down.
- $(x - p)$ shifts right, $(x + p)$ shifts left.
- $-f(x)$ reflects in the x -axis, $f(-x)$ reflects in the y -axis.
- $a \cdot f(x)$ stretches vertically.

Average gradient between a and b :

- $(f(b) - f(a)) \div (b - a)$.

Practice questions

1. Given $f(x) = (x - 2)^2 - 9$, write down the turning point and the x-intercepts.
2. Sketch $y = 2^x - 4$. State the asymptote, the y-intercept and the range.
3. For $y = -3/(x + 1) + 2$, write down the asymptotes, the domain and the range.
4. The parabola $y = a(x + p)^2 + q$ has turning point $(-1 ; 4)$ and passes through $(0 ; 1)$. Find the value of a .
5. Calculate the average gradient of $f(x) = x^2 - 2x$ between $x = 1$ and $x = 4$.
6. State the domain and range of $y = 1/(x - 3) - 2$.
7. For $f(x) = 5 \times 2^x$, find $f(-1)$.
8. The graph of $y = x^2$ is shifted 3 units to the right and 2 units up. Write the new equation.

Answers

1. Turning point $(2 ; -9)$. For the x-intercepts, set $y = 0$: $(x - 2)^2 = 9$, so $x - 2 = \pm 3$, giving $x = 5$ or $x = -1$.
2. Asymptote $y = -4$. y-intercept: $2^0 - 4 = 1 - 4 = -3$, so the point is $(0 ; -3)$. Since $a > 0$, the range is $y > -4$.
3. Asymptotes $x = -1$ (vertical) and $y = 2$ (horizontal). Domain: $x \neq -1$. Range: $y \neq 2$.
4. With turning point $(-1 ; 4)$, the form is $y = a(x + 1)^2 + 4$. Substitute $(0 ; 1)$: $1 = a(1)^2 + 4$, so $a = -3$. The equation is $y = -3(x + 1)^2 + 4$.
5. $f(4) = 16 - 8 = 8$ and $f(1) = 1 - 2 = -1$. Average gradient $= (8 - (-1)) \div (4 - 1) = 9 \div 3 = 3$.
6. Domain: $x \neq 3$. Range: $y \neq -2$.
7. $f(-1) = 5 \times 2^{-1} = 5 \times 1/2 = 2.5$.
8. A right shift of 3 gives $(x - 3)^2$, and an upward shift of 2 adds 2, so the new equation is $y = (x - 3)^2 + 2$.

Official EduRank Team notes. Always cross-check with your textbook and teacher.